Pre Calculus 12

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1 Transformations

1.1 Functions and Their Graphs

Relations:

- A relation R is a set of ordered pairs
- The domain of R is the set of first coordinates of the ordered pairs
- the range R is the set of second coordinates of the ordered pairs

A relation can be represented in different ways:

- a set of ordered pairs
- a table of values
- a graph
- an equation

y is equal to the $\sqrt{x}$ where x is all integers that are perfect squares between (0, 16)

x	y
y = 0	x = 0
y = 1	x = 1
y = 2	x = 4
y = 3	x = 9
y = 4	x = 16

for: $y = 2x + 1$
Domain = [0, -2, -4, 2, 4]
Range = [1, -3, -7, 5, 9]

What is the purpose of joining the points with a line?
- joining the points shows the relations include more ordered pairs not just the ones for this set of values

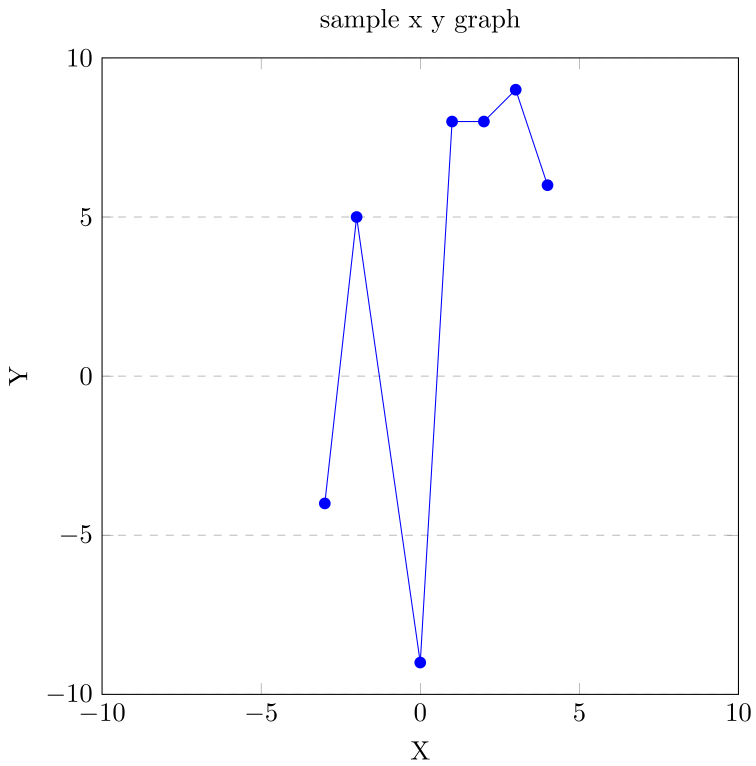
Conventions:
- no endpoints means no termination for graph
- open dot means approaches and gets infinitely close
- always read inequalities from the variable to the number $x \leq 7$ or $7 \leq x$ is "x is greater than or equal to 7"

Relation as expressed by x,y graph:
Domain: set of x values represented by the graph
- always start at left most point and trace to right most point
- $-6 < x \leq 7$ (here -6 is an open dot and 7 is a closed dot so we can get to 7 but we can never get to -6)
Range: set of y values represented by the graph
- can start at max or min
- $-4 \leq y < 7$
Domain: $-3 \leq x \leq 8$
Range: $-2 \leq y \leq 8$

What do functions do?
- for every possible input value there is one output value
- a function is a special kind of relation
- there is a difference between functions and relations

Defintion of a function:
- a function is a relation which assigns to each element in the domain(x) at single element in the range(y)
- for every value of the domain there is one and only one value of the range
- if two points can be joined by a vertical line then the graph is not a function

Is an equation a function?
- if an x value can be found that produces more than one value of y then the equation is not a function
- $y = x^2$ (if an equation is solved for y it is a function)
- $y^2 = x$ (here to simplify for y we get $y = +/-\sqrt{x}$ so there are two values of y for x ($x = 4 \therefore y = 2 \ \& \ y = -2$))
- if y^2 or if $x = \mathbb{R}$ it's not a function



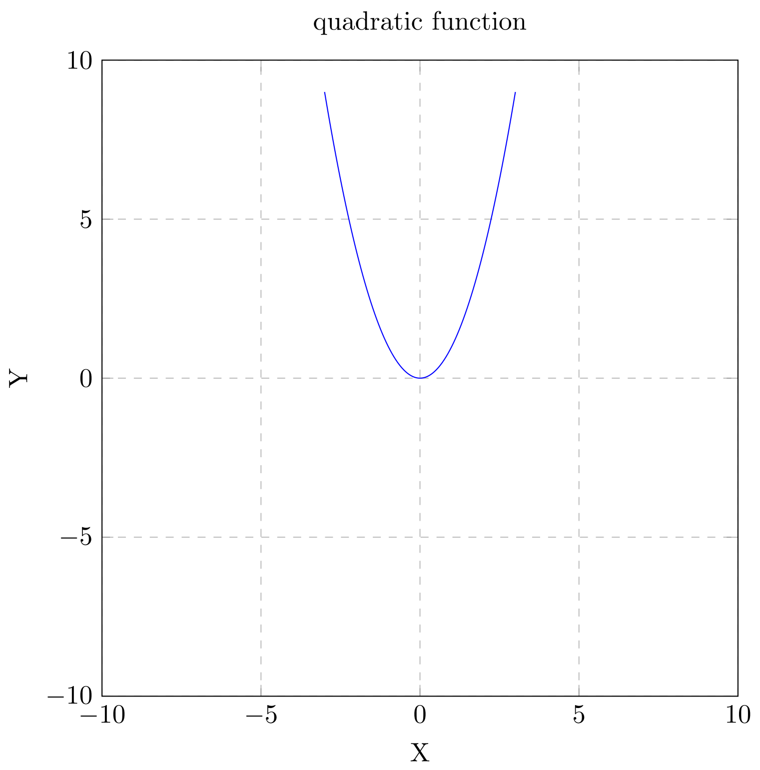
X + Y Intercept:
$2y + 3x = -6$
- to find y intercept make $x = 0$ in function $\therefore y = -3$
  $2y + 3(0) = -6$
  $\frac{y}{2} = \frac{-6}{2}$
  $y = -3$
- to find x intercept make $y = 0$ in function $\therefore x = -2$
  $2(0) + 3x = -6$
  $\frac{3x}{3} = \frac{-6}{3}$
  $x = -2$

Slope Intercept Form:
$y = mx + b$
slope $= \frac{rise}{run}$
only rise or run can be -ve
  $2y = -6 - 3x$
  $y = \frac{-6-3x}{2}$
  $y = \frac{-3}{2}x - 3$

The Constant Function:
$y = 1$
$y = mx + b$ aka The Slope Intercept form
The domain for a constant f is all $\mathbb{R}$
The range is a single $\mathbb{R}$ like 1

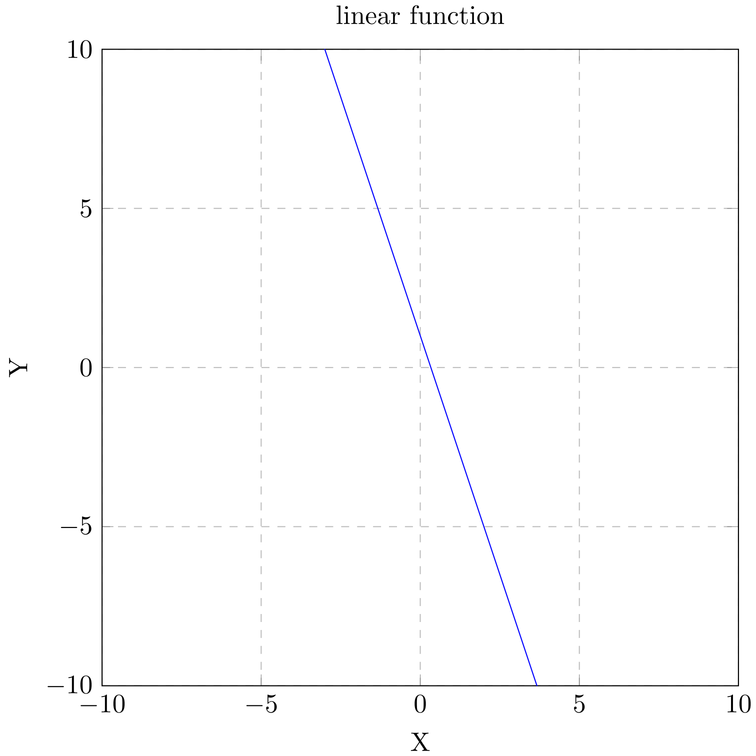
The Quadratic Function:
$y = x^2$

x	y
x = 3	y = 9
x = 2	y = 4
x = 1	y = 1
x = 0	y = 0
x = -1	y = 1
x = -2	y = 4
x = -3	y = 9

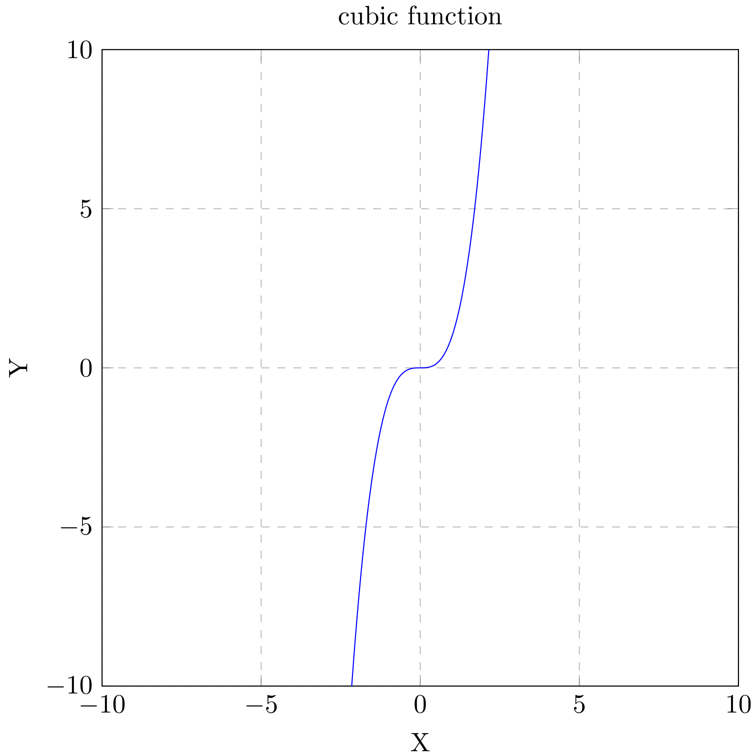


$D : x \in \mathbb{R}$
$R : y \geq 0$
- every parabola has an axis of symettry where $x = 0$
- every vertical line the equation will always be x = "where the line crosses 0"

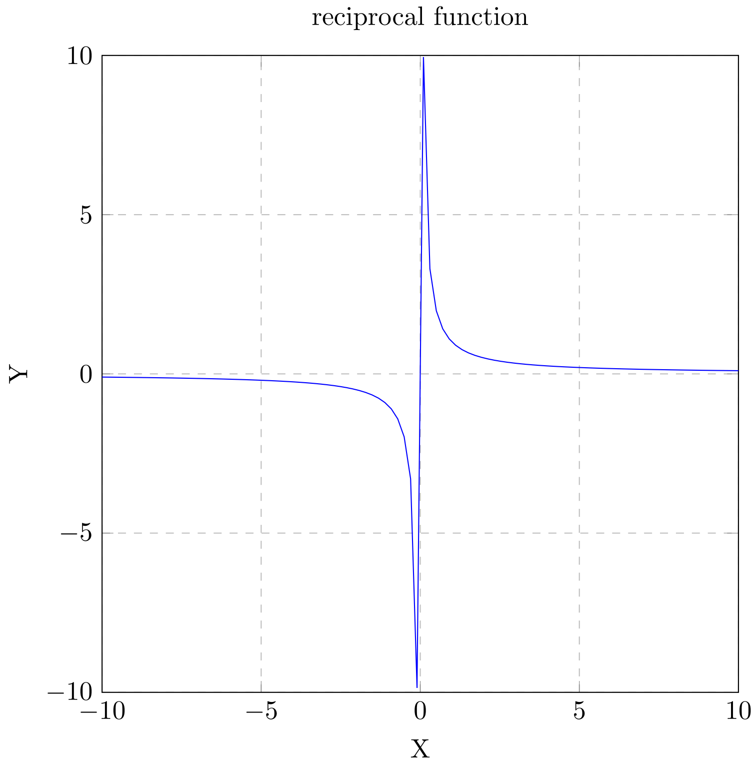
Linear Function:
$y = x$
$y = -3x + 1$
$D : x \in \mathbb{R}$
$R : y \in \mathbb{R}$



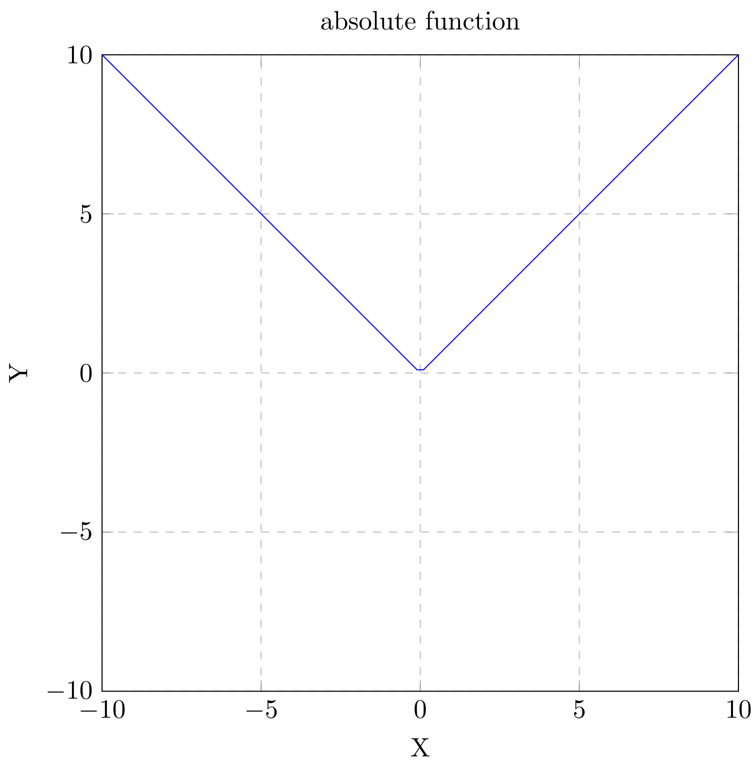
Cubic Function:
$y = x^3$
$D : x \in \mathbb{R}$
$R : y \in \mathbb{R}$
- results in an S shaped curve



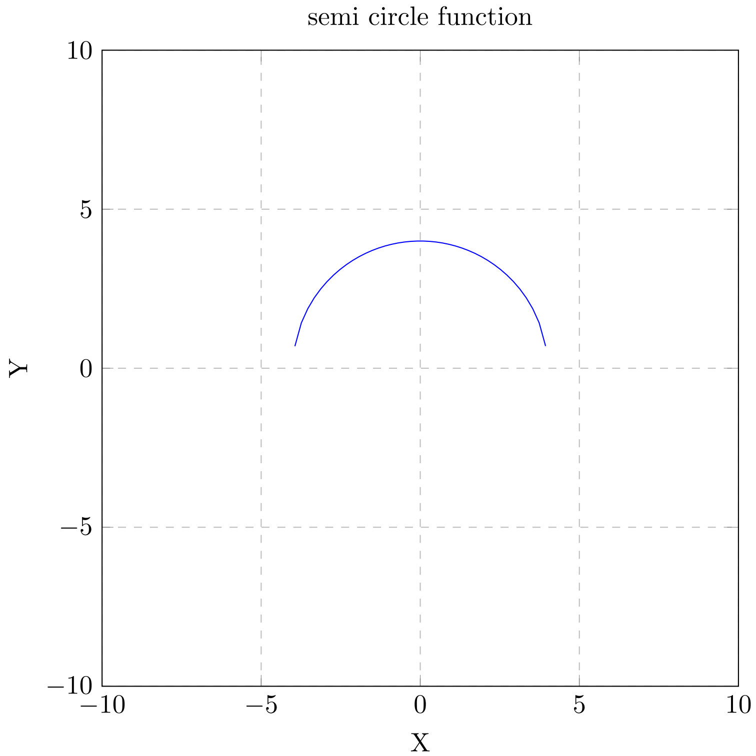
Reciprocal Function:
$y = \frac{1}{x}$
$D : x \neq 0 \ \ \& \ \ x \in \mathbb{R}$
$R : y \neq 0 \ \ \& \ \ y \in \mathbb{R}$
- Asymptote: is a line where the graph will approach the line infinitely but never touch the line



Absolute Function:
$y = |x|$
$D : x \in \mathbb{R}$
$R : y \geq 0$

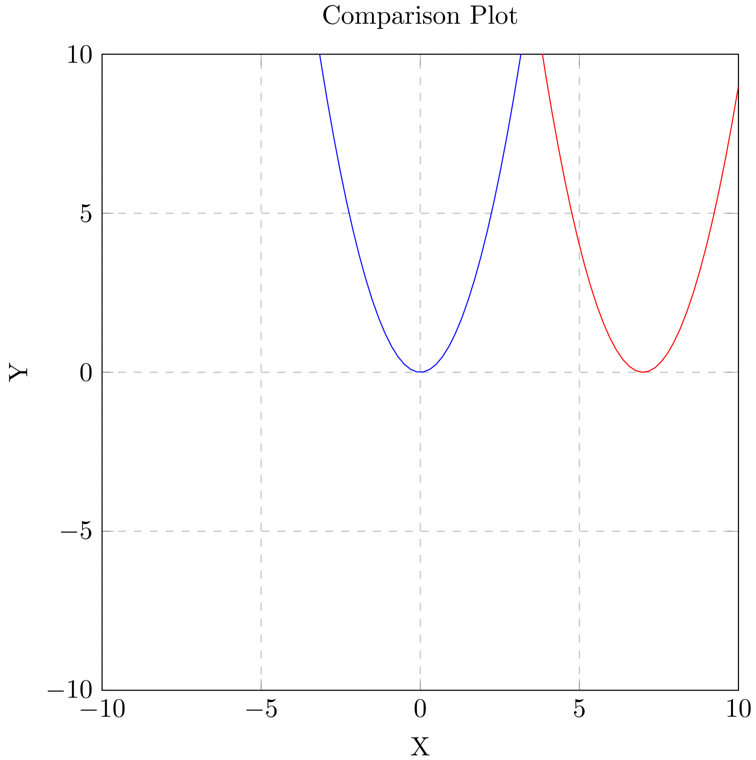


Semi Circle Function:
$y = \sqrt{16 - x^2}$
$D : -4 \leq x \leq 4$
$R : 0 \leq y \leq 4$
- $\sqrt{16}$ gives us the domain and the range

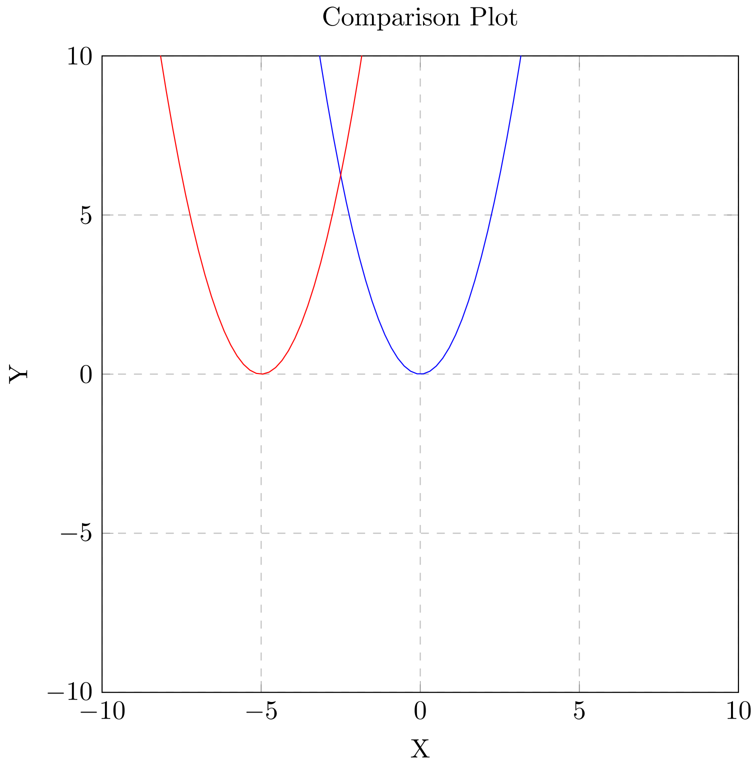


1.2 Translating Graphs of Functions

Comparing $y = x^2$ **to** $y = (x - 7)^2$:
- replacing x with x - h - moves the graph by a translation (7 units to the right)



Comparing $y = x^2$ **to** $y = (x + 5)^2$:
- replacing x with x - h
- moves the graph by a translation (5 units to the left)

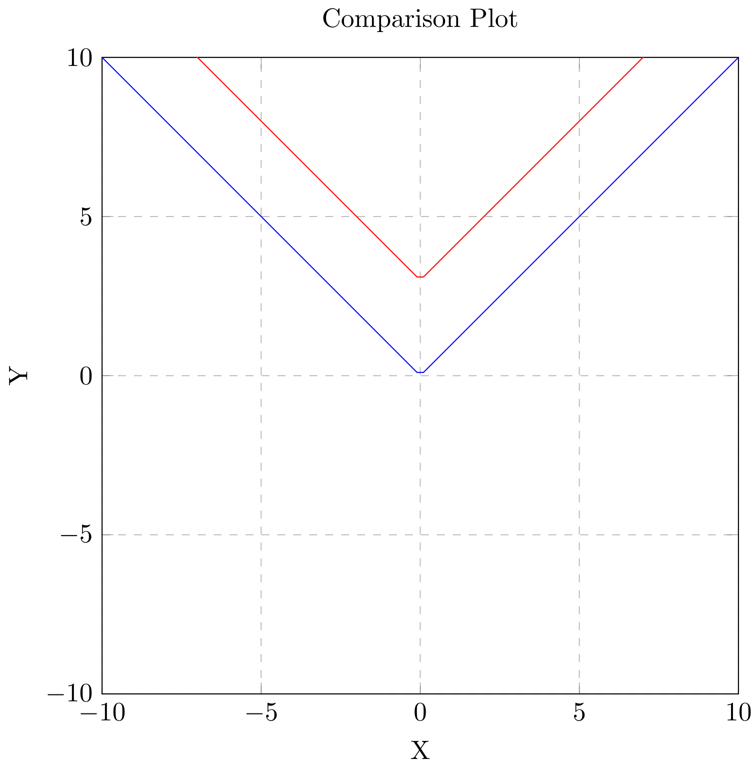


Comparing $y = f(x)$ **to** $y = f(x - h)$:
- so $y = x^2$ becomes $y = (x - h)^2$
- h refers to horizontal translation

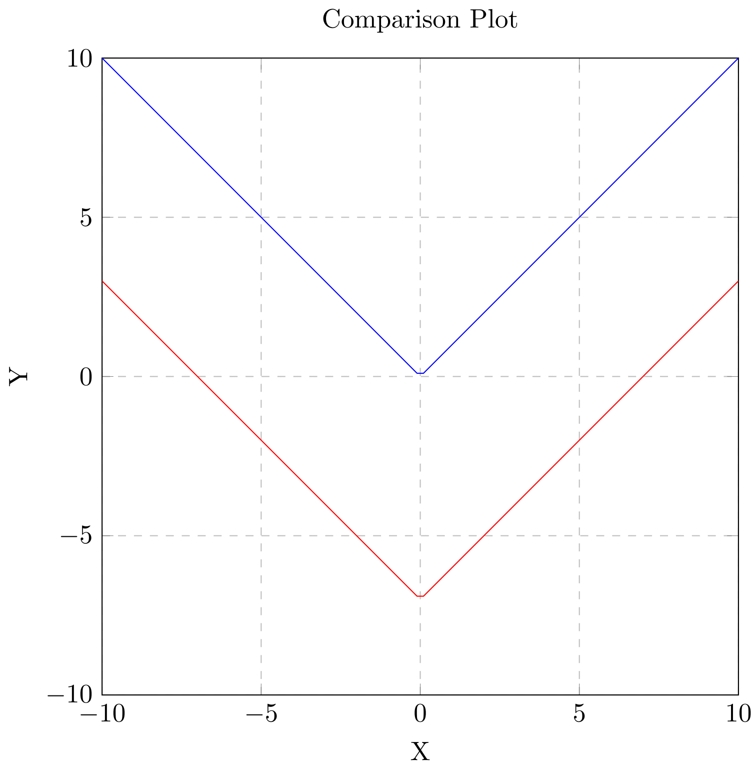
- if $h > 0$ then the graph is translated h units to the right
- if $h < 0$ then the graph is translated h units to the left
- if function is absolute ie $y = |x|$
- $y = |x - 5|$ translates the graph 5 units to the right
- in this case $h = 5$
- if h is +ve like $y = |x + 3|$ then translate 3 units to the left
- in this case $h = -3$

- write the new equation for each translation on the function $y = \sqrt{x}$
a) a horizontal translation of 5 units left
- $y = \sqrt{x + 5} \therefore h = -5$
b) a horizontal translation of 3 units right
- $y = \sqrt{x - 3} \therefore h = 3$

Comparing $y = |x|$ to $y - 3 = |x|$:
- replacing y with $y - k$
- k is vertical translation
- $k = 3$ and the graph is translated 3 units up vertically
- $y - 3 = |x|$ is equal to $y = |x| + 3$



Comparing $y = |x|$ to $y + 7 = |x|$:
- replacing y with $y - k$
- k is vertical translation
- $k = -7$ and the graph is translated 7 units down vertically
- $y + 7 = |x|$ is equal to $y = |x| - 7$


Comparison Plot
10
5
0
−5
−10
Y
−10
−5
0
5
10
X

Comparing $y = f(x)$ to $y + k = f(x)$:
- k refers to vertical translation
- if $k < 0$ then the graph is translated k units up
- if $k > 0$ then the graph is translated k units down

Example 01: - Sketch the graph of the function $y = \sqrt{x+3} - 4$
a) What is the domain and the range of this new function
D: $x \geq -3$
R: $y \geq -4$
b) what are the x and y intercepts?
X intercept:
$y = 0$
$0 = \sqrt{x+3} - 4$
$4 = \sqrt{x+3}$
$4^2 = (\sqrt{x+3})^2$
$16 = x + 3$
$x = 13$
Y intercept:
$x = 0$
$y = \sqrt{0+3} - 4$
$y = \sqrt{3} - 4$

Example 02: - Sketch the graph of the function $y = (x-4)^2 + 1$
a) What is the domain and the range of this function?
D: $x = \mathbb{R}$
D: $y \geq 1$
b) what are the x and y intercepts?
X intercept:
$y = 0$
$0 = (x-4)^2 + 1$
$-1 = (x-4)^2$
- we cannot do $\sqrt{-1}$ $\therefore$ no x intercept
Y intercept:
$x = 0$
$y = (0-4)^2 + 1$
$y = (-4)^2 + 1$
$y = 16 + 1$
$y = 17$

Example 03:
- Sketch the graph of the function $y = \sqrt{16 - (x-3)^2}$
a) What is the domain and the range of this function?
$D: -1 \leq x \leq 7$ this is a translation the -3 becomes the h value 3 so offset by +3
$R: 0 \leq y \leq 4$

Example 04:
- Given the graph of the function $y = f(x)$ sketch the graph $y = f(x+3) - 2$
- graph is translated 3 units to the left $x + 3$
- graph is tranlated 2 units down -2

Example 05:
- Given the point $(3, -4)$ determine the new coordinates of the translated image $y = f(x+4) + 2$
- translate 4 to the left (-4)
- translate 2 up, (+2)
- $(-1, -2)$

Formative Assessment:

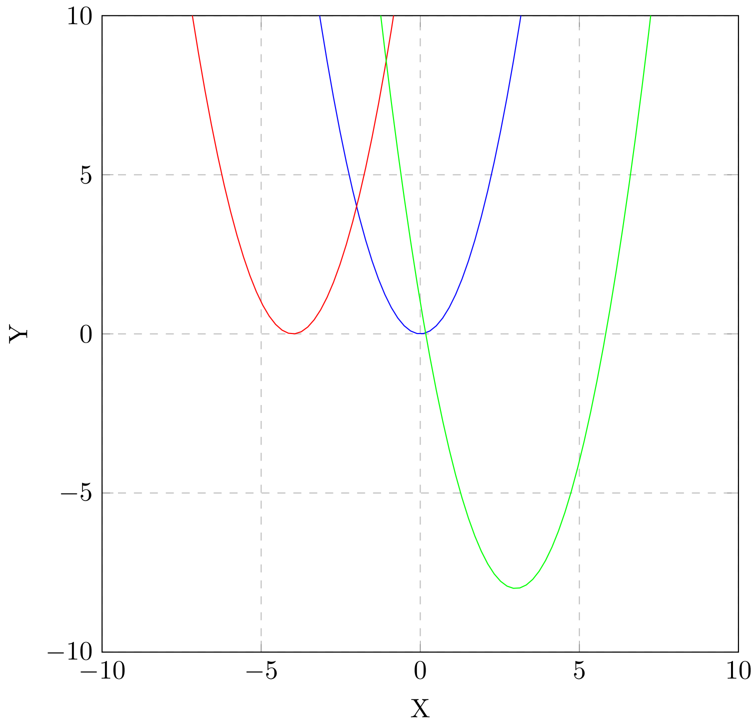
1) Given the graph of the function $y = f(x)$ sketch the graph $y + 4 = f(x + 3)$
   $y = f(x + 3) - 4$
   translate 3 units to the left and 4 units down

2) Write the equation of the image of $y = f(x)$ after each transformation
   a) a horizontal translation of 9 units left
      $y = f(x + 9)$
   b) A vertical translation of 3 units up
      $y = f(x) + 3$
   c) A horizontal translation of 1 unit left and 7 units down
      $y = f(x + 1) - 7$

3) What happens to the graph of the function $y = f(x)$ if you make the following changes
   a) replace x with $x - 6$
      translate 6 units to the right
   b) replace y with $y + 4$
      translate 4 units down
   c) replace x with $x + 2$ and y with $y - 5$
      translate 2 units to the left and 5 units up

4) Sketch the graphs of the function on the same grid
   a) $y = x^2$
   b) $y = (x + 4)^2$
   c) $y = (x - 3)^2 - 8$
      these are all parabolic functions with translations

Comparison Plot



5) The function $y = f(x)$ is to tranformed to $y = f(x - c) + d$ determine the value of c and d for each
   TAKE NOTE OF THE - BEFORE C WHEN SOLVING THIS, LEFT IS $y = f(x + 3)$ BUT IT BECOMES -3 TO GET -(-3) or +3
   a) 3 units left $\therefore c = -3$ & $d = 0$
   b) 2 units down $\therefore c = 0$ & $d = -2$
   c) 6 units right and 9 units down $\therefore c = +6$ & $d = -9$

6) Sketch the graphs of the function on the same grid

a) $y = \sqrt{4 - x^2}$

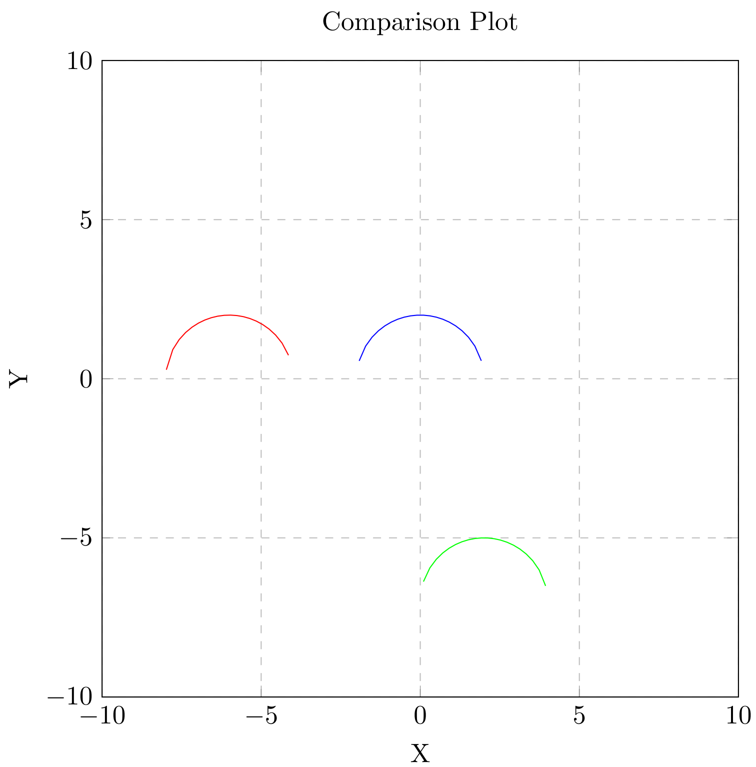
b) $y = \sqrt{4 - (x + 6)^2}$

translated 6 units to the left

c) $y = \sqrt{4 - (x - 2)^2} - 7$

translated 2 units to the right and 7 units down

These are all semi circle functions with translations



7) Sketch the graphs of the function on the same grid

a) $y = \frac{1}{x}$

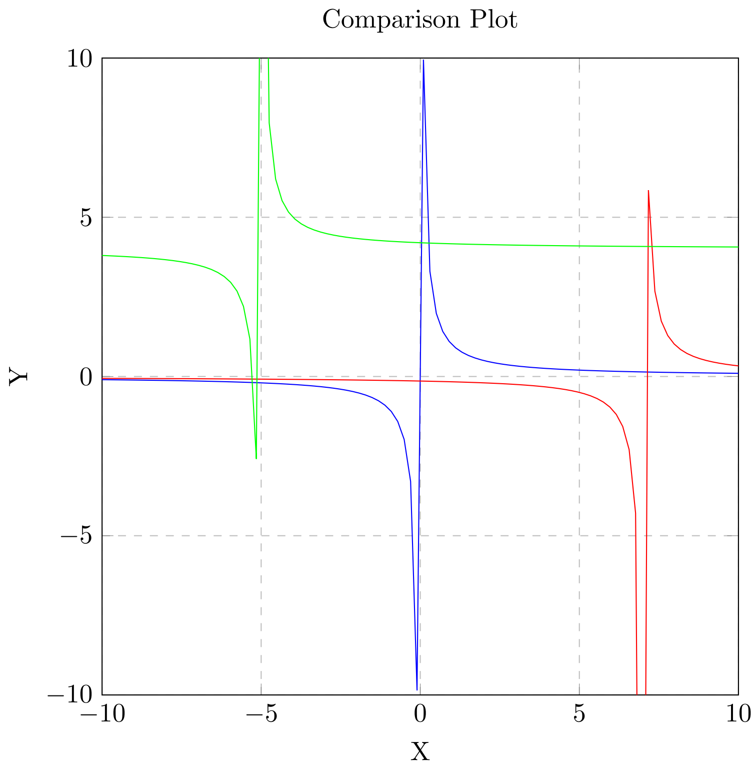
b) $y = \frac{1}{x-7}$

translated 7 units to the right

c) $y = \frac{1}{x+5} + 4$

translated 5 units to the left and 4 units up

These are all reciprocal functions with translations



8) The function represented by the solid line is a transformation of the given function. Write an equation for the transformation

a) $y = |x|$

4 units to the right and 6 units down

$y = |x - 4| - 6$

b) $y = \sqrt{x}$

4 units to the left and 6 units up

$y = \sqrt{x + 4} + 6$

9) Give the graph of the function $y = f(x)$ sketch the graph of $y = f(x + 5)$

translate graph 5 units to the left

10) Give the graph of the function $y = f(x)$ sketch the graph of $y - 6 = f(x)$

becomes $y = f(x) + 6$

translate graph 6 units up

11) Give the graph of the function $y = f(x)$ sketch the graph of $y = f(x - 3) + 2$

translate graph 3 untis to the right and 2 units up

Quiz Stuff:

Question 3: Determine the domain and range of the function $y = \frac{1}{x-2}$

my answer: $D : x \neq 2 \ \& \ R : y \in \mathbb{R}$

correct answer: $D : x \neq 2 \ \& \ R : y \neq 0$

- this is reciprocal function so both $y \neq 0 \ \& \ y \in \mathbb{R}$ but here the $y \neq 0$ is correct

Question 3: if (a, b) is a point on the graph $y = f(x)$ dethermine a point on the graph of $y = f(x - 2) + 3$

my answer: $(a - 2, b + 3)$

correct answer: $(a + 2, b + 3)$

- $y = f(x - 2) + 3$ is a translation of 2 units to the right and 3 units up so the a value is + 2

THIS IS SUPER CONFUSING SO MAKE SURE YOU WALK THROUGH THIS TRANSLATION WHEN ANSWERING THESE QUESTIONS

1.3 REFLECTING GRAPHS OF FUNCTIONS

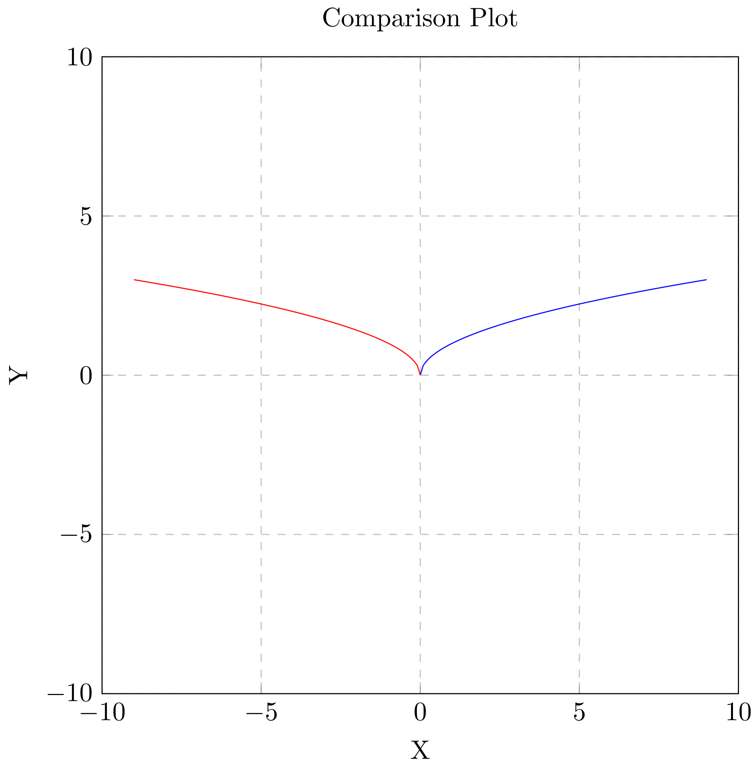
Comparing $y = \sqrt{x}$ to $y = \sqrt{-x}$:

$y = \sqrt{x}$ Base Points

x	y
x = 0	y = 0
x = 1	y = 1
x = 4	y = 2
x = 9	y = 3

$y = \sqrt{-x}$
- you can not take the square root of a negative number $\therefore x \neq$ a positive number
- x can only be negative numbers and zero

x	y
x = 0	y = 0
x = -1	y = 1
x = -4	y = 2
x = -9	y = 3



Desribe the effect of the graph when $y = \sqrt{x}$ is replaced by $y = \sqrt{-x}$:
- the graph is reflected in the y axis
- the point $(0, 0)$ has a special name, it's called an **INVARIANT** point
- **INVARIANT POINT: A point that is unchanged by a transformation**

Comparing $y = \sqrt{x}$ to $-y = \sqrt{x}$:

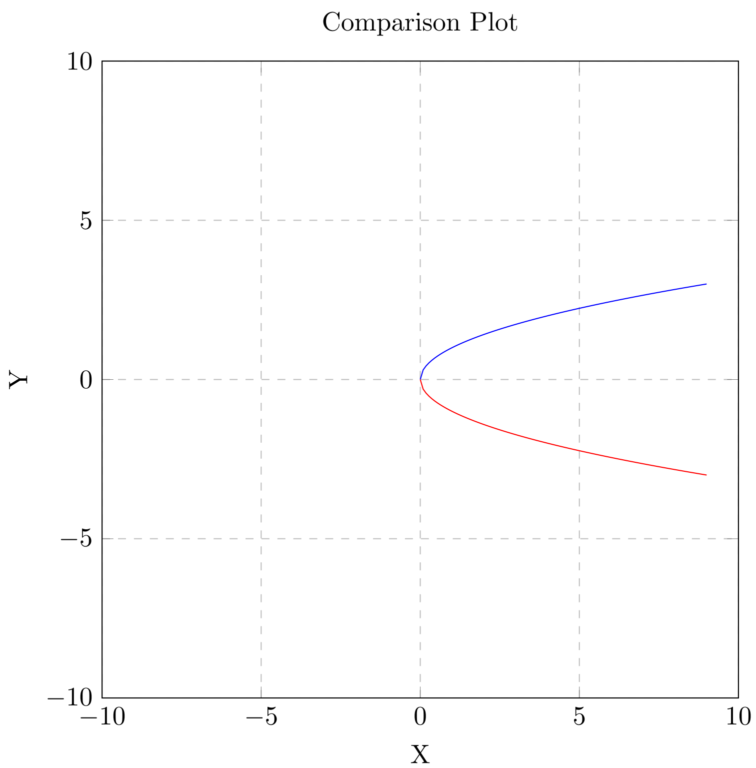
$y = \sqrt{x}$ Base Points

x	y
x = 0	y = 0
x = 1	y = 1
x = 4	y = 2
x = 9	y = 3

$y = \sqrt{-x}$
IMPORTANT: Solve for y first

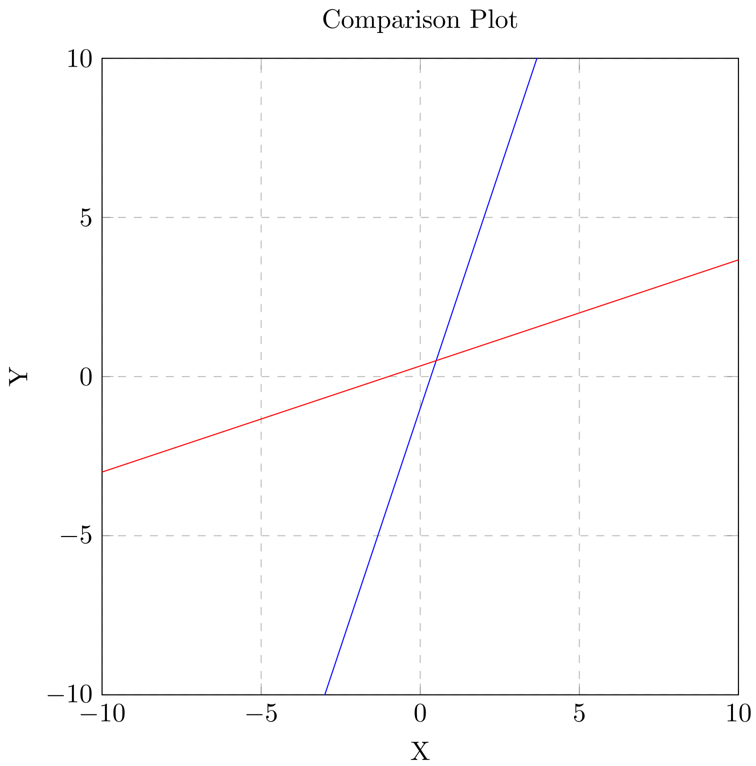
$-y = \sqrt{x}$
$\frac{-y}{-1} = \frac{\sqrt{x}}{-1}$
$y = -\sqrt{x}$
$y = -\sqrt{1}$
$y = -1$

x	y
x = 0	y = 0
x = 1	y = -1
x = 4	y = -2
x = 9	y = -3



Desribe the effect of the graph when $y = \sqrt{x}$ is replaced by $-y = \sqrt{x}$:
- the graph is reflected in the x axis
- $-y = \sqrt{x}$ is the same as $y = -\sqrt{x}$
- this reflection also has an invariant point at (0,0)

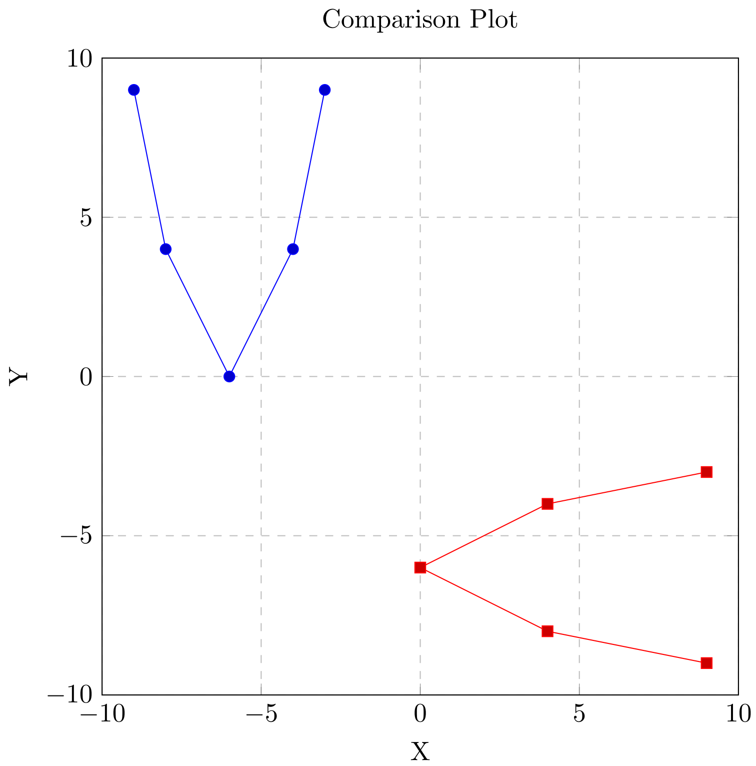
Comparing $y = f(x)$ to $x = f(y)$:
$x = f(y)$ this is the **INVERSE** of $y = f(x)$
Graph the linear function $y = 3x - 1$ (Slope intercept form)
-1 is the y intercept and the $\frac{rise}{run}$ is $\frac{3}{1}$
slope is +ve so line goes from lower left to upper right
Graph the inverse function $x = 3y - 1$ (swap x and y)
Solve for y
$x + 1 = 3y$
$\frac{x}{3} + \frac{1}{3} = \frac{3y}{3}$
$y = \frac{1}{3}x + \frac{1}{3}$



Desribe the effect of the graph when $y = 3x - 1$ when x and y are switched :
- switching x and y describes a REFLECTION in the line $y = x$
NOTE: $y = f^{-1}(x)$ is sometimes used to represent the the equation of the inverse
f^{-1} is NOT a negative exponent, it simply means the inverse

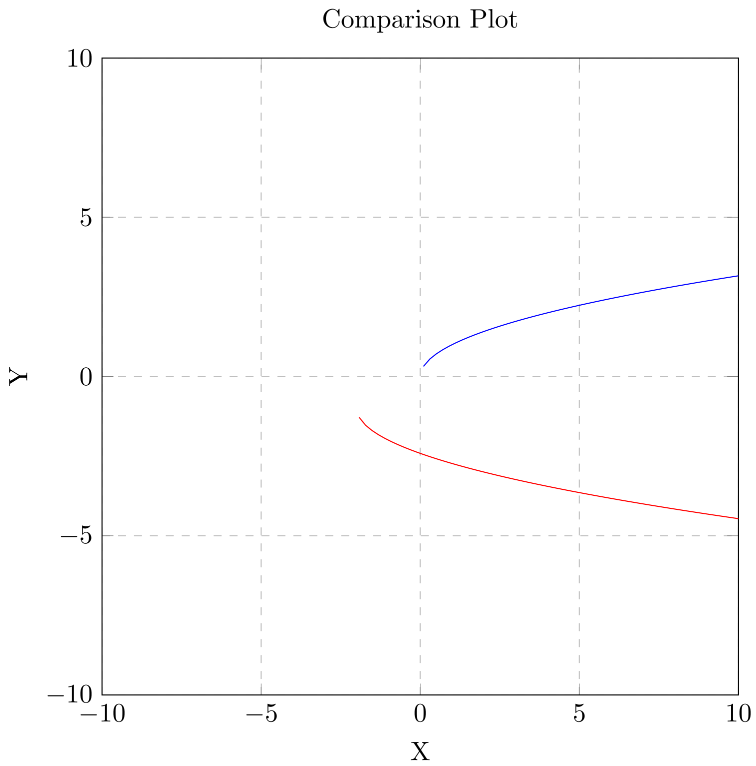
- reflecting points in this Comparison
- the inverse is not a function, it's a relation, not a function
- you can use the method of reflecting base points on the line y = x to graph the inverse but it is not - to graph the inverse use the method of switch x, y

$y = f(x)$	$y = f^{-1}(x)$
(-3, 9)	(9, -3)
(-4, 4)	(4, -4)
(-6, 0)	(0, -6)
(-8, 4)	(4, -8)
(-9, 9)	(9, -9)

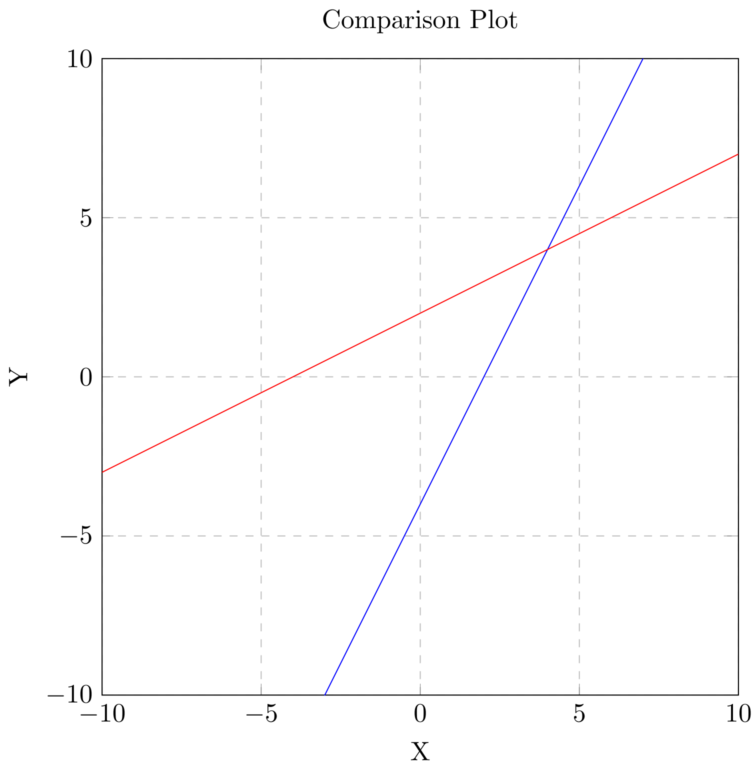


Examples:
1) Graph the function $y = -\sqrt{x+2} + 1$
- this is a reflection on the x axis
- solve for y
Steps from solution:
Step 1) dethermine the base equation and then graph the base function
the base equation is $y = \sqrt{x}$
Step 2) Identify the transformations of the function
the -ve value out the fucntion indicates a reflection on the x axis
the x+2 indicates a shift of 2 units to the left
the -1 indicates a shift of 1 unit down

x	y
0	-2
2	-3
7	-4
14	-5



2) Graph the function $f(x) = 2x - 4$ then sketch the graph of $y = f^{-1}(x)$ (the inverse)
You should identify the function as a linear function
The graph is a straight line
To graph locate the y intercept (-4) and the slope (2)
so rise/run is $\frac{2}{1}$
draw the slope from the y intercept
Now that you have the graph drawn choose base points
Then switch the x and y values



3) Determine $y = f^{-1}(x)$ the inverse of $f(x) = \frac{1}{x-2}$
$y = \frac{1}{x-2}$ so the inverse is $x = \frac{1}{y-2}$
solve for y
multiply each side by the common denominator $y - 2$
$(y-2)x = \frac{1}{y-2}(y-2)$
Cancel the $y - 2$
$(y-2)x = 1$
Divide both sides by x
$\frac{(y-2)x}{x} = \frac{1}{x}$
cancel the x's
$y - 2 = \frac{1}{x}$
$y = \frac{1}{x} + 2$

4) Given the graph of the function $y = f(x)$ below, graph $y = -f(x)$
$y = -f(x)$ indicates a reflection on the x axis
base points of the original function are

x	y
-7	0
-4	3
3	-3
6	7

x values remain constant and y values are inverted

x	y
-7	0
-4	-3
3	3
6	-7

5) Given the graph of the function $y = f(x)$ below, graph $y = f(-x)$
$y = f(-x)$ indicates a reflection on the y axis

x	y
-7	0
-4	3
3	-3
6	7

y values remain constant and x values are inverted

x	y
7	0
4	3
-3	-3
-6	7

6) Given the graph of the function $y = f(x)$ below, graph $y = f^{-1}(x)$
$y = f^{-1}(x)$ indicates an inverse function

x	y
-7	0
-4	3
3	-3
6	7

invert the x and y values

x	y
0	-7
3	-4
-3	3
7	6

7) If the point $(-3, 2)$ is on the graph of the function $y = f(x)$ determine the new coordinates of the image of this point on the function
a) $y = f(-x)$
reflection on the y axis, x value is inverted so $(3, 2)$
b) $y = -f(x)$
reflection on the x axis, y value is inverted so $(-3, -2)$
c) $y = f^{-1}(x)$
inversion, x and y value is inverted so $(2, -3)$

formative Assessment relflecting graphs of functions